

Width-2 coverability mechanism behind the unilateral strip obstruction

To force one vertex, the target needs a diagonal antisymmetric component $A_j(1,-1)$.
A seedless column only carries $A_j(v_j)$, where v_j is the rung label.

symmetric rail mode

fresh width-2 column
no seeds inside

single rung

antisymmetric mode

Because every nonzero label satisfies $a+b \neq 0$, v_j is never parallel to $(1,-1)$.
The missing diagonal component blocks the first activation in a seedless column.